Answer Key with Explanations

ANSWER KEY WITH EXPLANATIONS

SAT-Style Practice Test — Full-Length Adaptive Edition

Total Questions: 147

Section	Module	Questions
S1 Reading & Writing	Module 1	27
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	TOTAL	**147**

SECTION 1: READING AND WRITING

Module 1 — Answer Key with Explanations

Q1: B “Gravity” in context means seriousness or solemnity. The passage describes a “quiet gravity” that connects personal identity to historical lineage — a mood of deep, solemn reflection, not physical heaviness (A), gravitational attraction (C), or downward movement (D).

Q2: C “Press” in context means to push physically. The passage describes sunlight pushing against Meursault’s eyes “with a physical force.” It does not mean to publish (A), to urge or persuade (B), or to iron clothing (D).

Q3: A “Permissible” means allowed but not required, which fits the context perfectly: discoverers *may* name comets after themselves but are “not obligated to do so.” “Infeasible” (B) means not possible. “Selective” (C) means careful in choosing. “Mandatory” (D) means required — the opposite of what the passage conveys.

Q4: B “Palpable” means readily perceived or easily noticeable. The influence is described as something visitors “can readily identify,” making “palpable” the best fit. “Corroborated” (A) means confirmed by evidence. “Disputed” (C) means challenged. “Understated” (D) means downplayed — contradicting the idea that the influence is easy to observe.

Q5: D The underlined sentence generalizes the gentleman’s behavior into a category of travelers who “approach foreign resorts as conquerors,” directly illustrating his condescending attitude toward the other people in the dining room. It does not describe eagerness to be noticed (A), a desire to own a restaurant (B), or a reason travelers avoid resorts (C).

Q6: C The underlined portion explains that “the complexity of the ending is only an issue” if the books are treated as independent — thereby offering a reason not to consider the complex ending a flaw. It does not argue about critical acclaim (B), reading order preferences (A), or engagement levels (D).

Q7: A The underlined portion states that “extensive travels abroad” sparked Ando’s engagement with sacred spaces. This is biographical information that accounts for the career transition from urban residential projects to contemplative designs. It does not describe departures from typical style (B), support the minimalism argument (C), or explain peer reception (D).

Q8: C The text states that Morales honors both historical figures (Cáceres) and unrecognized individuals (the fisherman), viewing both as “representing the resilience and interconnectedness of all communities.” This is captured by choice C. The text does not describe a career transition (A), a specific focus on communal bonds over hardship (B), or comparative scholarly attention (D).

Q9: D The passage’s central question is why some frogs call and others use visual displays, and the answer offered is structural differences in the vocal sac. Choice D captures this main idea. Choice A is a supporting detail. Choice B is too narrow. Choice C mischaracterizes the researchers’ position as uncertain.

Q10: A The text uses Kahneman’s dual-process theory to *explain* Mehta et al.’s finding about branded apps. The theory provides a framework: cognitive load from app features leaves no capacity for marketing messages. Choice B overstates the causal link. Choice C contradicts the text. Choice D misattributes the conclusion to Kahneman rather than Mehta et al.

Q11: B The table directly shows the Zune (manufactured by Microsoft) sold approximately 3,200,000 units. Choice A (14,500,000) is the iPod Classic. Choice C (2,200,000) is the MiniDisc Walkman. Choice D (500,000) is the Diamond Rio.

Q12: B The statement needs to complete: “the global increase occurred even though ______.” This requires a contrast — a country whose production *decreased*. Norway went from 1.14 to 0.66 billion barrels, a decrease. Choice A is factually wrong (Saudi Arabia produced over 3 billion in 2000). Choice C describes an increase, not a contrast. Choice D is factually wrong (Norway produced less than 1 billion in 2020).

Q13: C The claim is that Pip’s attendance at the grand dinner reflects his “self-conscious ambition.” Choice C directly describes Pip’s “restless aspiration” that made sitting at a fine table feel “both thrilling and agonizing” because of his humble origins — perfectly illustrating self-conscious ambition. Choice A addresses social fiction, not personal ambition. Choice B is about children and injustice. Choice D is about shame, not aspiration.

Q14: D The hypothesis is that smart bins reduce *collection frequency* more effectively than they reduce *overall waste volume*. Choice D directly supports this: smart bins produced “significantly reduced collection trips but only marginal decreases in total waste tonnage.” Choices A and C deal with MWEI scores without distinguishing between frequency and volume. Choice B contradicts the expected benefit.

Q15: A Smartphones use lithium-ion batteries (cobalt >= 15%, greater energy density). Household electronics use low-self-discharge nickel-metal hydride batteries (lower cobalt, lower energy density). Therefore, smartphone batteries have greater energy density. Choice B reverses the cobalt logic. Choice C attributes crystalline structure to lithium-ion, but the text says crystalline structure belongs to nickel-metal hydride. Choice D ignores that the two battery types have *different* electrode chemistry.

Q16: B “Daniela Gutierrez-Pham, a transportation planner and cycling advocate,” is a nonessential appositive phrase that requires commas on both sides to properly set it off from the sentence. Choice A lacks both commas. Choice C lacks the opening comma. Choice D lacks the closing comma.

Q17: D “Sites that have their own historic stone circles” is a restrictive (essential) clause that should not be enclosed in commas. A period then starts a new sentence: “An enormous stone circle…” Choice A uses commas around a restrictive clause and creates a run-on. Choices C and D also mishandle the clause boundary.

Q18: A A period after “stations” creates two grammatically complete sentences: “…number nearly one hundred and fifty stations. Located across the globe, the seismometers record earthquake activity…” A comma (B) would create a comma splice. Choice C creates a run-on relative clause. Choice D fuses two independent clauses.

Q19: C The subject is “Catalyst” (singular), which requires the singular verb “couples.” “Couple” (A) is plural. “Are coupled” (B) is plural. “Have coupled” (D) is plural. Only “couples” (C) is the correct singular present tense.

Q20: B This sentence lists three people in a series with internal commas (each name has an appositive). Semicolons must separate the items: “Rosalind Franklin, British biophysicist of the Cambridge School; Gerardus Mercator, Flemish cartographer…” Choice A uses commas where semicolons are needed. Choice C places the semicolon after the wrong element. Choice D omits necessary punctuation.

Q21: D A period after “mechanism” creates a valid new sentence: “Now widely understood to serve functions…the phenomenon provides similar advantages…” Without the period, choices A, B, and C all create comma splices or run-on sentences joining two independent clauses.

Q22: D “Later” indicates chronological sequence — Carson’s 1962 serialization came first, and other environmental works followed in subsequent years (1971, 1989). “That is” (A) introduces a restatement. “Previously” (B) reverses the chronology. “For example” (C) would introduce an instance of the same event.

Q23: B “In turn” indicates a consequential relationship: the increased vigor of sodium’s reaction *led to* a rise from moderate to extreme explosive tendency. “In sum” (A) means in total. “Specifically” (C) would introduce a narrower example. “For example” (D) would introduce an illustration rather than a consequence.

Q24: C “For example” introduces a specific instance (the Arctic tern’s 44,000-mile migration) that illustrates the general claim about migratory birds traveling longer distances. “However” (A) would introduce a contrast. “Meanwhile” (B) implies simultaneous events. “Additionally” (D) would add a new claim rather than illustrate one.

Q25: A To contrast the two sites, we need details showing how they differ. Choice A directly contrasts their construction settings: one beneath a rock overhang on the side of a canyon, the other on an open plain. Choice B reverses the facts. Choice C claims both are no longer inhabited (contradicted by the notes for Taos Pueblo). Choice D incorrectly implies only one was Ancestral Puebloan.

Q26: A The goal is to emphasize a *similarity*. Choice A highlights what the two novels share: both are Booker Prize winners. Choice B mentions only one novel. Choice C compares the novelists rather than the novels. Choice D contrasts rather than emphasizes similarity.

Q27: D The goal is to describe the *role* of heating systems in greenhouses. Choice D directly explains: greenhouses use heating systems like gas-fired boilers to maintain optimal growing temperatures by burning fuel. Choice A omits the mechanism. Choice B describes propane’s composition, not the system’s role. Choice C incorrectly states energy converts “back.”

Module 2 Easier — Answer Key with Explanations

Q1: C “Continuity” means an unbroken connection over time. Scholars trace thematic links from oral traditions to modern literature — a continuous thread. “Subtlety” (A) refers to delicacy. “Sophistication” (B) refers to complexity. “Innovation” (D) refers to novelty, not persistence.

Q2: D “Irrefutable” means impossible to dispute. The reasons for characterizing Monet as Impressionist “may seem” beyond argument, yet the characterization is too narrow. “Incongruous” (A) means out of place. “Pretentious” (B) means pompous. “Provocative” (C) means deliberately controversial.

Q3: B “Manifest” in this context means perceptible or noticeable — the river is “visible” in silvery flashes. “Situated” (A) means located. “Realized” (C) means achieved. “Dwindling” (D) means shrinking.

Q4: D “Palpable” means readily perceived. The Mughal influence is easily “observable” throughout the subcontinent. “Corroborated” (A) means confirmed. “Disputed” (C) means challenged. “Understated” (B) means downplayed.

Q5: B The underlined sentence elaborates on what biased analysis (from small sample size) can lead to: “reports of exaggerated effects.” It is a consequence of the limitation described in the previous sentence. It does not summarize a shift (A), emphasize the effect's magnitude (C), or counter Reid's objection (D).

Q6: B The three underlined portions — walking the same path for ten years, carrying her basket home like a soldier from a campaign, and sitting heavily feeling defeated — collectively suggest exhaustion and weariness with an oppressively routine life. They do not convey nervous tension (A), a husband's expectations (C), or a specific failed errand (D).

Q7: C The text describes the Welwitschia species and explains how it developed varied survival strategies through diversification from a single ancestral lineage adapting to extreme aridity. It does not claim all Namib plants are Welwitschia (A), identify ancestors of all desert plants (B), or identify other species sharing an ancestor (D).

Q8: B Text 1 presents the problem of inconsistent definitions of food insecurity across countries. Text 2 describes a standardized framework (the Integrated Food Security Phase Classification) that addresses precisely this problem. The author of Text 2 would most likely note that a possible solution is available (B).

Q9: A Okafor and Li reconstructed provincial revenue records that "demonstrate that Akbar's periodic treasury shortfalls were caused by disruptions in provincial remittance cycles, *not* systemic administrative corruption." They used new evidence to eliminate one explanation (corruption) and affirm another (remittance disruptions). They did not reveal how shortfalls caused unrest (B), correct Habib and Richards's errors (C), or support Habib and Richards's hypothesis (D).

Q10: B The second team claims orchids use scents "primarily to attract specific pollinator species." If orchids also produce scent during encounters with *herbivorous* insects (B), scent would serve a purpose beyond pollinator attraction — directly challenging the claim. Choice A actually supports the claim. Choice C is tangential. Choice D describes an acoustic property unrelated to the claim.

Q11: A The research question is whether *perceived category similarity* predicts likelihood of purchasing brand extensions. Choice A tests this directly: have households rate the similarity of product categories, then correlate those ratings with calculated purchase probabilities. Choice B tests brand recognition vs. cost. Choice C tests within-category similarity. Choice D tests brand recognition vs. frequency.

Q12: A If P4 gave *equal* ratings to STEM and Humanities courses, but the model's correlation is lower for STEM (0.35) than Humanities (0.42), the model's predictions must have *differed* from P4's actual ratings by different amounts in each subject. Therefore, the model predicted different ratings (A). Choice B assumes a directional conclusion not supported. Choice C contradicts the different correlations. Choice D reverses the direction.

Q13: A The researchers' caution against generalizing stems from their sample being exclusively from South Korea, a "highly competitive streaming market." To determine whether the findings apply broadly, research is needed with participants from countries with varying levels of market competition (A). Choice B makes an unsupported prediction. Choice C overgeneralizes. Choice D suggests a larger sample from the same country, missing the point about geographic limitation.

Q14: D Two-host zooplankton are “typically shielded from thermal stress by their host organisms,” yet their abundance decreased while directly exposed zooplankton remained stable. This suggests that whatever advantage the host-mediated protection offered, it was not sufficient to offset other negative temperature-driven effects (D). Choice A reverses the finding. Choice B extends beyond the data. Choice C contradicts the passage.

Q15: A The passage states that for “every class of animals examined, there were individual studies showing differences well above the average for birds.” Since mammals are a class, some individual mammal studies must have shown effects above the bird average. This means some mammal studies found larger effects than some bird studies (A). Choice B introduces a detail not discussed. Choice C reverses the stated relationship. Choice D makes an unsupported claim about Park’s specific study.

Q16: B The sentence requires parallel verb forms: “exported textiles…and ______ other luxury goods.” Since “exported” is simple past tense, the parallel structure requires “imported” (B). “To import” (A) is infinitive. “Importing” (C) is a participle. “Having imported” (D) is a perfect participle.

Q17: D The National Museum of Norway is a singular institution, requiring the singular possessive pronoun “its” (no apostrophe). “Their” (A) is plural. “It’s” (B) means “it is.” “His” (C) is masculine, inapplicable to a museum.

Q18: C A colon introduces an elaboration of what the spacecraft detected: “the observation of a vast nitrogen-ice plain.” The colon correctly links the general discovery to the specific details. A period (A) would break the logical connection. “It was the” (B) creates a comma splice. A comma (D) creates a comma splice with two independent clauses.

Q19: C “However” introduces a new independent clause that contrasts with the preceding sentence. Two independent clauses require a period (or semicolon) between them, not a comma. “Surface. However,” (C) correctly separates the two sentences. Choice A misuses a semicolon-however-comma combination. Choice B creates a comma splice. Choice D places the semicolon after “however” incorrectly.

Q20: C “Compounds glucose and methanol” is an essential (restrictive) appositive identifying *which* compounds. Essential appositives take no commas: “The liquid compounds glucose and methanol have molar masses…” Choices A and B incorrectly add commas around a restrictive appositive. Choice D adds a trailing comma with no opening comma.

Q21: C A period after “tenets” creates two complete sentences: “…central tenets. Developed by Italian physician Maria Montessori…the method emphasizes learning…” Without the period, choices A and B create comma splices joining two independent clauses. Choice D is ungrammatical.

Q22: D “By contrast” signals a comparison between the two photographers’ different contributions: Bourke-White for photojournalism, Parks for photography *and* filmmaking. “As a result” (A) implies causation. “For instance” (B) implies illustration. “In other words” (C) implies restatement.

Q23: B “That is” introduces a restatement or definition — here, defining “semantic drift” as “the rate at which a word’s meaning will shift over time.” “Likewise” (A) draws a parallel. “For example” (C) introduces an illustration. “In addition” (D) adds new information.

Q24: D “Accordingly” means “as a natural consequence.” The general pattern is that regional crafts were displayed as limited-run demonstrations. Accordingly, the specific example follows logically: ikebana appeared only once, while glassblowing (being internationally recognized) became permanent. “Moreover” (A) simply adds. “Granted” (C) concedes a counterpoint. “However” (B) introduces a contrast.

Q25: A The goal is to emphasize a *difference*. Choice A identifies the specific difference: *November Steps* features a biwa (lute) while *A Flock Descends* incorporates a shakuhachi (flute). Choice B states a shared attribute. Choice C emphasizes similarity. Choice D is too general.

Q26: A The goal is to *define* “high-sinuosity river.” Choice A provides the definition: a river with many meanders (U-shaped curves). Choice B adds information about change over time, going beyond a definition. Choice C incorrectly states that oxbow lakes *cause* sinuosity. Choice D describes the Mississippi specifically rather than defining the term.

Q27: A The goal is to *contrast* the two orbits. Choice A contrasts both shape (elliptical) and period (3.55 vs. 7.15 Earth days). Choice B merely lists facts about Kepler’s laws. Choice C states that Europa orbits faster but omits specific data. Choice D emphasizes similarity, not contrast.

Module 2 Harder — Answer Key with Explanations

Q1: B “Coherence” means logical consistency and connectedness. Despite geographic dispersion, artistic traditions display “striking thematic” coherence — recurring motifs connecting diverse cultures. “Ambiguity” (A) means vagueness. “Spontaneity” (C) means impulsiveness. “Uniformity” (D) implies identical sameness rather than thematic interconnection.

Q2: C “Incontrovertible” means impossible to deny or dispute. The basis for classifying Kahlo as Surrealist “may seem” irrefutable, yet the label is overly reductive. “Incongruous” (A) means out of place. “Derivative” (B) means unoriginal. “Unsubstantiated” (D) means unsupported — contradicting “may seem.”

Q3: A “Permissible” means allowed but not required. Experimentalists *may* adjust dosage protocols “but they are not compelled to do so.” “Unprecedented” (B) means never done before. “Compulsory” (C) means mandatory — the opposite. “Negligible” (D) means insignificant.

Q4: D “Conspicuous” means easily noticed or prominent. The Renaissance influence is something visitors “can readily identify” throughout European institutions. “Understated” (A) means downplayed. “Corroborated” (B) means confirmed. “Disputed” (C) means challenged.

Q5: D The sentence generalizes the officer's behavior: administrators who "regard colonial outposts as personal dominions" inspire "obedience without loyalty." This illustrates his condescending attitude toward the people at the outpost. It does not explain administrative difficulty (A), characterize the porters' reaction (B), or evaluate managerial effectiveness (C).

Q6: C The underlined portion explains that narrative fragmentation is "only disorienting if one reads the novels as self-contained works" — offering a framework for understanding why the fragmentation should not be considered a flaw. It does not compare critical acclaim (A), rank accessibility (B), or identify the most complex novels (D).

Q7: A The underlined portion states that "an extended period of philosophical immersion" catalyzed Borges's turn toward metaphysical fiction. This is biographical information that accounts for the career transition from lyric poetry to fiction. It does not note departures from typical style (B), support the lyric aesthetic argument (C), or explain peer praise (D).

Q8: B Anatsui's installations honor both historically significant events (the slave trade in *Dusasa*) and everyday resourcefulness (the cloth merchant's stall), viewing both as "representing the resilience and interconnectedness of West African communities." Choice A makes an unsupported claim about scholarly attention. Choice C describes an unsupported career trajectory. Choice D mischaracterizes the focus.

Q9: A Arrow's concept of moral hazard provides the explanatory framework: employees who receive wellness benefits may reduce their own health-maintenance efforts, offsetting intended gains. This explains why wellness programs fail to reduce costs despite increasing participation. Choice B reverses the causal claim. Choice C contradicts the text. Choice D misattributes the conclusion.

Q10: B The second team claims nutrient sharing occurs "primarily between parent trees and their offspring as a form of kin selection." If the highest-transfer pairs were *unrelated species* (B), nutrient sharing would not be limited to kin — directly challenging the claim. Choice A addresses network density, not kinship. Choice C describes size differences, not kinship. Choice D supports the claim rather than challenging it.

Q11: D The research question is whether *perceived mission alignment* predicts consumers' willingness to pay a *price premium*. Choice D tests this directly: have households rate mission alignment, then correlate those ratings with the calculated price premium. Choice A tests brand awareness vs. cost. Choice B tests quality vs. income. Choice C tests frequency vs. price.

Q12: C If P3 reported *identical* engagement scores for both modalities, but the model's correlation differs (0.40 In-Person vs. 0.52 Online), the model must have predicted *different* engagement ratings for the two modalities. Therefore the model predicted that P3's ratings would differ (C). Choice A and D assume directionality unsupported by correlation data alone. Choice B contradicts the different correlations.

Q13: A The researchers' caution against generalizing stems from their sample being exclusively from France, which has a "highly developed art-house film culture." To determine whether findings apply broadly, research is needed with participants from countries with varying levels of art-house culture (A). Choice B makes an unsupported prediction. Choice C overgeneralizes. Choice D misidentifies the limitation as sample size rather than geographic scope.

Q14: D Two-host branchiopods are “typically shielded from osmotic stress by their host environments,” yet their abundance decreased while directly exposed branchiopods remained stable. Whatever advantages host-mediated protection offered, they did not fully offset other salinity-driven negative effects (D). Choice A reverses the finding. Choice B extends beyond the data. Choice C contradicts the passage.

Q15: B For “every phylum of organisms examined, there were individual studies showing differences well above the average for corals.” Since sponges are a phylum, some individual sponge studies must have shown effects above the coral average — meaning some sponge studies found larger effects than some coral studies (B). Choice A introduces an irrelevant criterion. Choice C reverses the average relationship. Choice D makes an unsupported claim.

Q16: C A colon after “archives” introduces the elaboration of how the library expanded: “its curators acquired rare manuscripts, early printed books, and cartographic materials.” The colon correctly links the general statement to specific details. “Archives, and” (A) creates a faulty compound. “Archives and” (B) creates a run-on. “Archives, its curators acquired” (D) creates a comma splice.

Q17: D “Roger Angell’s reputation as one of the greatest baseball writers of all time,” followed by a comma, sets up a long nonessential modifier (“bolstered by dozens of *New Yorker* essays…”) before reaching the main verb later. A comma after “time” properly marks the start of this parenthetical modifier. Choices A and C incorrectly introduce a verb (“was”/“had been”) that would derail the sentence structure. Choice B misplaces the comma.

Q18: A The parenthetical “for example” must be set off by commas on both sides when embedded in a sentence. “1890, for example,” (A) correctly places commas before and after the parenthetical, then continues with “while.” Choice B misuses a semicolon. Choice C misplaces the semicolon after “example.” Choice D lacks the opening comma.

Q19: B A colon after “1648” introduces an elaboration explaining the *result* of the separation: “the latter’s connection to the empire is so attenuated that Spanish is seldom spoken there today.” The colon links the historical fact to its consequence. “And” (A) treats the clauses as equal rather than explanatory. A comma (C) creates a comma splice. No punctuation (D) creates a run-on.

Q20: C The subject of the sentence is the gerund phrase “Distinguishing ‘exactly the same’ and ‘nearly the same’ commodities.” A gerund phrase functioning as a subject is singular, requiring the singular verb “was.” “Having been” (A) is participial, not a finite verb. “Being” (B) is also participial. “Were” (D) is plural.

Q21: D This sentence uses a complex series with internal commas (each item has a subordinate clause). Semicolons must separate the major items: “*The Siege of Thebes*…pilgrims’ return; *The Tale of Beryn*…tourists; and *The Ploughman’s Tale*…” “Return” (A) omits the semicolon. “Return,” (B) uses a comma where a semicolon is needed. “Return.” (C) would prematurely end the sentence.

Q22: B “By contrast” signals a comparison between the two photographers’ different subjects: Lange documented displaced farmworkers; Evans studied rural Southern architectural structures. “As a result” (A) implies causation. “In other words” (C) implies restatement. “For instance” (D) would introduce an example rather than a contrast.

Q23: A “That is” introduces a restatement or definition — here, defining “connotative shift” as “the rate at which a word’s implied associations will evolve over time.” “For example” (B) introduces an illustration. “Likewise” (C) draws a parallel. “In addition” (D) adds new information.

Q24: C “Accordingly” means “as a natural consequence.” The general pattern is that regional crafts appeared as temporary side exhibitions. Accordingly, the specific example follows logically: mask-making appeared only once, while glass sculpture (internationally celebrated) became permanent. “Moreover” (A) simply adds. “Granted” (B) concedes a counterpoint. “However” (D) introduces a contrast.

Q25: B The goal is to highlight a *difference*. Choice B identifies the specific contrast: the Church of the Light uses a cruciform aperture for natural sunlight, while the Water Temple uses a lotus pond as its entrance requiring visitors to descend through water. Choice A emphasizes similarity (both minimalist concrete). Choice C also emphasizes similarity. Choice D states chronology, not difference.

Q26: A The goal is to *define* “high-eruptive-frequency volcano.” Choice A provides the definition: one that erupts many times over a geologically short period, producing calderas and pyroclastic deposits. Choice B compares two volcanoes rather than defining the term. Choice C confuses location with definition. Choice D describes a single volcano.

Q27: D The goal is to *contrast* the two orbits. Choice D effectively identifies the key difference: despite sharing prograde direction and near-circular shape, Titan’s orbital period (15.95 days) is substantially longer than Enceladus’s (1.37 days). Choice A emphasizes similarities. Choice B mixes orbital data with unrelated facts. Choice C also emphasizes similarity.

SECTION 2: MATH

Module 1 — Answer Key with Explanations

Q1: C Each output is 52 times the previous: 1/52 x 52 = 1, 1 x 52 = 52, 52 x 52 = 2,704. A constant multiplicative ratio indicates an increasing exponential function with base 52.

Q2: A g(-4) = -5(-4) + 30 = 20 + 30 = 50.

Q3: 25 x^2 = (25)(25) = 625. The positive solution is x = sqrt(625) = 25.

Q4: C Both data sets have 11 values, so the median is the 6th value when sorted. Data set A (sorted): 3, 3, 3, 6, 6, **9**, 12, 12, 15, 15, 15. Median = 9. Data set B (sorted): 3, 6, 6, 9, 9, **9**, 9, 12, 12, 15, 15. Median = 9. The medians are equal.

Q5: 35 $f(3) = 3^3 + 8 = 27 + 8 = 35$.

Q6: B Translating a graph down by k units subtracts k from the output: $y = 7^x - 5$. Left/right shifts affect the exponent (C and D), not the output.

Q7: 8 Set the equations equal: $4x + 12 = (x + 2)^2 = x^2 + 4x + 4$. Simplify: $12 = x^2 + 4$, so $x^2 = 8$.

Q8: C $f(2) = 120$ means: when $x = 2$ (i.e., 2 seconds after launch), the height $f(x)$ is 120 feet. This matches choice C. Choice A reverses x and $f(x)$. Choice B claims a maximum without justification. Choice D contradicts $f(2) = 120$ (not zero).

Q9: A In the equation $8x + 12y = 296$, the coefficient 12 is the price per adult ticket, and y is the quantity multiplied by that price. Therefore, y represents the number of adult tickets sold.

Q10: B The total solute from both solutions equals the solute in the mixture: (concentration_1)(volume_1) + (concentration_2)(volume_2) = (concentration_mix)(total_volume). This gives $0.04x + 0.10y = 0.06(x + y)$.

Q11: 24 In a right triangle, $\sin(x)$ = opposite/hypotenuse. Angle x is opposite the side of length 24, so $\sin(x) = 24/c$. Therefore $k = 24$.

Q12: 6 Reading directly from the bar graph, the bar at data value 2 reaches a height (frequency) of 6.

Q13: D The scale factor between similar triangles = $252/36 = 7$. Since AB corresponds to DE and AB = 12, we have DE = $12 \times 7 = 84$.

Q14: 11 The equation $(x + 3)^2 + (y + 5)^2 = 121$ is in center-radius form with $r^2 = 121$. Therefore $r = \sqrt{121} = 11$.

Q15: C Set up the system: $x + y = 84$ and $x = 3y + 10$. Substituting: $3y + 10 + y = 84$, so $4y = 74$ and $y = 18.5$. Then $x = 3(18.5) + 10 = 65.5$.

Q16: D Let w = width. Then length = w + 8. Area: $w(w + 8) = 180$. Expanding: $w^2 + 8w - 180 = 0$. Factoring: $(w + 18)(w - 10) = 0$. Since width must be positive, $w = 10$.

Q17: A First 30 hours at $40/hr: 30 x 40 = $1,200. Remaining needed: $1,900 - $1,200 = $700. Additional hours at $60/hr: 700/60 = 11.67, rounded up to 12. Total hours: 30 + 12 = 42. Verify: 41 hours gives $1,200 + 11(60) = $1,860 < $1,900. So 42 is the minimum.

Q18: A The graphs of the linear and quadratic equations intersect at approximately $x = 2$ and $x = 8$. Among the choices, $x = 2$ (choice A) is a valid intersection point.

Q19: A Distributing: $13x - 156 = -15x - 195$. Combining: $28x = -39$, so $x = -39/28$. This is exactly one solution.

Q20: 263 Using elimination on $13x - 5y = 26$ and $4x + 3y = 37$: Multiply the first equation by 3: $39x - 15y = 78$. Multiply the second by 5: $20x + 15y = 185$. Add: $59x = 263$, so $x = 263/59$. Therefore $p = 263$.

Q21: 206/9 Find the slope: $m = (-179/10 - (-197/10)) / (3 - 1) = (18/10) / 2 = 9/10$. Using point $(1, -197/10)$: $y = (9/10)(x - 1) + (-197/10) = (9/10)x - 206/10$. Setting $y = 0$: $(9/10)x = 206/10$, so $x =$

206/9.

Q22: D Since LM is parallel to PQ, triangles LMR and QPR are similar (by AA similarity: alternate interior angles and vertical angles at R). The ratio LR/QR = MR/PR gives $8/QR = 6/15$, so $QR = (8 \times 15)/6 = 20$. Therefore $LQ = LR + QR = 8 + 20 = 28$.

Module 2 Easier — Answer Key with Explanations

Q1: C $36 = 8 + 4(x - 2)$. Isolating: $4(x - 2) = 36 - 8 = 28$.

Q2: D $(x^2 + 9)^2 + (x - 7)(x + 7) = x^4 + 18x^2 + 81 + x^2 - 49 = x^4 + 19x^2 + 32$.

Q3: B Since the graph passes through (7, 11), by definition $f(7) = 11$.

Q4: C At $x = 0$: $y = 8(3)^{(0+2)} = 8(3^2) = 8(9) = 72$. The y-intercept is (0, 72).

Q5: D Check each table against $y > 3x - 4$. Table B: when $x = 2$, $y = 5 > 2$; when $x = 4$, $y = 11 > 8$; when $x = 7$, $y = 20 > 17$. All values satisfy the inequality. Other tables have at least one value that fails: e.g., Table C has $y = 17$ when $3(7) - 4 = 17$, but 17 is not greater than 17.

Q6: B Slope $= (194 - 128)/(6 - 4) = 66/2 = 33$. Using point (4, 128): $p = 33w + b$, so $128 = 33(4) + b$, giving $b = -4$. The equation $p = 33w - 4$ can be rewritten as $33w - p = 4$.

Q7: D $h(8) = 3(7(8) - 8) = 3(56 - 8) = 3(48) = 144$.

Q8: -1/3 The line $y = x + 1/3$ crosses the x-axis when $y = 0$: $0 = a + 1/3$, so $a = -1/3$.

Q9: B $x^2 + 4x = 60$. Completing the square: $x^2 + 4x + 4 = 64$, so $(x + 2)^2 = 64$. Then $x + 2 = \pm 8$, giving $x = -2 + 8 = 6$ or $x = -2 - 8 = -10$. Choice B: $-2 + \sqrt{64} = -2 + 8 = 6$.

Q10: C Data set P is data set Q with every value increased by 9 (1->10, 2->11, etc.). Adding a constant to every data point shifts the mean but does not change the spread. Therefore the standard deviations are equal: $r = t$.

Q11: D The ratio $7/x = y/6$. Cross-multiplying: $7 \times 6 = xy$, so $42 = xy$, giving $x = 42/y$.

Q12: A Points decrease by 25% each 4-second interval, meaning 75% is retained. After x seconds (x a multiple of 4): $f(x) = 360(0.75)^{(x/4)}$. Choice B uses 4x instead of x/4 in the exponent. Choices C and D use 0.25 (the decrease rate, not the retention rate).

Q13: C The number of 7-inch bolts = 3n, the number of 5-inch bolts = n, and the number of 2-inch bolts = 20. Total count (not weight): $3n + n + 20 = 4n + 20 = 80$. Choice A incorrectly multiplies by bolt size. Choice B omits the given 20 small bolts. Choice D omits one group.

Q14: B The radius from center (3, 4) to tangent point (a, -2) is perpendicular to the tangent line (slope 5/6). Perpendicular slope = -6/5. Radius slope: $(-2 - 4)/(a - 3) = -6/(a - 3)$. Setting equal to -6/5:

-6/(a - 3) = -6/5, so a - 3 = 5, and a = 8.

Q15: A The given leg runs from (-6, 2) to (6, 8). Its length = sqrt((12)^2 + (6)^2) = sqrt(180) = 6sqrt(5). Using the area formula: (1/2)(6sqrt(5))(other leg) = 36sqrt(5). Solving: other leg = 72sqrt(5) / (6sqrt(5)) = 12.

Q16: B We have PQ = ST = 8 and angle P = angle S = 40 degrees. The angle P is located at vertex P, between sides PQ and PR. If PR = SU (choice B), we have two sides and the included angle: PQ = ST, angle P = angle S, PR = SU. This is the SAS (Side-Angle-Side) congruence criterion. Choice A gives SSA (ambiguous case). Choice C (equal perimeters) is insufficient. Choice D is wrong because SSA alone does not guarantee congruence.

Q17: C The displayed graph of y = f(x) - 15 has a positive slope and crosses the y-axis above the origin. Since f(x) = y + 15, the function f also has a positive slope and a y-intercept shifted up by 15 (still positive). This matches f(x) = d + cx, where c (slope) and d (y-intercept) are positive constants.

Q18: 45/52 Non-Lepidoptera insects: Coleoptera (11 + 6 + 4 = 21) + Hymenoptera (8 + 20 + 3 = 31) = 52. Among these 52, those with wing length <= 25 mm: Coleoptera (11 + 6 = 17) + Hymenoptera (8 + 20 = 28) = 45. Probability = 45/52.

Q19: 111 Square both sides: k - x = (28 - x)^2 = 784 - 56x + x^2. So k = x^2 - 55x + 784. This quadratic in x has a minimum at x = 55/2 = 27.5. Minimum k = (27.5)^2 - 55(27.5) + 784 = 756.25 - 1512.5 + 784 = 27.75 = 111/4. Therefore 4k = 111. (Verify: at x = 27.5, sqrt(111/4 - 27.5) = sqrt(0.25) = 0.5, and 28 - 27.5 = 0.5. Checks out.)

Q20: D Test each choice by substituting x = -2b and checking whether the expression equals zero for some positive integer b. Choice D: 3(-2b)^2 + 25(-2b) + 14b = 12b^2 - 50b + 14b = 12b^2 - 36b = 12b(b - 3) = 0 when b = 3. Verify: 3x^2 + 25x + 42 = (3x + 7)(x + 6), and x + 6 = x + 2(3) = x + 2b. Choices A, B, and C yield no positive integer b.

Q21: 32 Rewrite: kx^2 - 36x + 10 = 0. For two distinct real solutions, the discriminant must be positive: (-36)^2 - 4(k)(10) > 0, so 1296 - 40k > 0, giving k < 32.4. The greatest integer value is k = 32.

Q22: 4 Mass of X = 320% of Y = 3.2Y. Mass of X = 0.080% of Z = 0.0008Z. Setting equal: 3.2Y = 0.0008Z, so Z = 3.2Y / 0.0008 = 4,000Y. Therefore Z is 4,000% of Y, meaning p = 4,000. And p/1,000 = 4.

Module 2 Harder — Answer Key with Explanations

Q1: C Translating y = 4^x left 3 units replaces x with (x + 3): y = 4^(x+3). Translating up 7 units adds 7: y = 4^(x+3) + 7. Choice A shifts right instead of left. Choice B shifts down instead of up. Choice D ignores the exponent shift.

Q2: D (x^2 + 7)^2 + (x - 5)(x + 5) = x^4 + 14x^2 + 49 + x^2 - 25 = x^4 + 15x^2 + 24. The first term expands using (a + b)^2 = a^2 + 2ab + b^2. The second term is a difference of squares.

Q3: B Set equal: 6x + 18 = (x + 3)^2 = x^2 + 6x + 9. Simplify: 18 = x^2 + 9, so x^2 = 9. (Solutions: x = 3 or x = -3. Check: when x = 3, y = 36; when x = -3, y = 0. Both satisfy y = (x + 3)^2.)

Q4: A For g(x) = -3x^2 + 54x - 192, the maximum occurs at the vertex: x = -b/(2a) = -54 / (2 x -3) = -54 / -6 = 9. The maximum value is g(9) = -3(81) + 54(9) - 192 = -243 + 486 - 192 = 51.

Q5: B Check each table against y > 4x - 7. Table B: when x = 2, y = 4 > 4(2) - 7 = 1; when x = 5, y = 16 > 4(5) - 7 = 13; when x = 9, y = 30 > 4(9) - 7 = 29. All values satisfy the inequality. Table A fails at x = 2: y = 1 is not strictly greater than 1. Table C fails at x = 9: y = 29 is not strictly greater than 29. Table D fails at x = 9: y = 29 is not strictly greater than 29.

Q6: C Slope = (201 - 123)/(5 - 3) = 78/2 = 39. Using point (3, 123): p = 39c + b, so 123 = 39(3) + b = 117 + b, giving b = 6. The equation p = 39c + 6 can be rewritten as p - 39c = 6. Verify with (8, 318): 39(8) + 6 = 318.

Q7: D f(-2) = 2(-2)^2 - 3(-2) + 1 = 2(4) + 6 + 1 = 15. f(1) = 2(1)^2 - 3(1) + 1 = 2 - 3 + 1 = 0. Therefore f(-2) - f(1) = 15 - 0 = 15.

Q8: -5/8 The line y = x + 5/8 crosses the x-axis when y = 0: 0 = a + 5/8, so a = -5/8.

Q9: B x^2 + 10x = 96. Completing the square: x^2 + 10x + 25 = 121, so (x + 5)^2 = 121. Then x + 5 = +/-11, giving x = -5 + 11 = 6 or x = -5 - 11 = -16. Choice B: -5 + sqrt(121) = -5 + 11 = 6.

Q10: C Data set R = {14, 14, 15, 15, 16, 17, 17, 18, 18, 18}. Data set S = {5, 5, 6, 6, 7, 8, 8, 9, 9, 9}. Each value in R is exactly 9 more than the corresponding value in S. Adding a constant shifts all values but does not change the spread (deviations from the mean remain identical). Therefore u = v.

Q11: D The ratio 11/x = y/8. Cross-multiplying: 11 x 8 = xy, so 88 = xy, giving x = 88/y.

Q12: A Points decrease by 30% every 3 seconds, meaning 70% is retained each interval. After x seconds (x a multiple of 3): f(x) = 800(0.70)^(x/3). Choice B uses 3x instead of x/3. Choices C and D use 0.30 (the decrease rate, not the retention rate).

Q13: C The number of large crates = 5n, the number of medium crates = n, and the number of small crates = 30. Total count: 5n + n + 30 = 6n + 30 = 120. Choice A incorrectly multiplies by crate size. Choice B counts only large and medium plus small. Choice D omits one group.

Q14: C The radius from center (7, 2) to tangent point (a, -5) is perpendicular to the tangent line (slope 4/3). Perpendicular slope = -3/4. Radius slope: (-5 - 2)/(a - 7) = -7/(a - 7). Setting equal to -3/4: -7/(a - 7) = -3/4. Cross-multiplying: -28 = -3(a - 7) = -3a + 21. So 3a = 49, and a = 49/3.

Q15: 12 The given leg runs from (-3, 1) to (5, 5). Its length = sqrt((8)^2 + (4)^2) = sqrt(64 + 16) = sqrt(80) = 4sqrt(5). Using the area formula: (1/2)(4sqrt(5))(other leg) = 24sqrt(5). Solving: other leg = 48sqrt(5) / (4sqrt(5)) = 12.

Q16: A We have MN = PQ = 10 and angle M = angle P = 55 degrees. Angle M is at vertex M, between sides MN and MO. If MO = PR (choice A), we have two sides and the included angle: MN = PQ, angle M = angle P, MO = PR. This is SAS (Side-Angle-Side) congruence. Choice B (equal perimeters) is insufficient. Choice C gives SSA (ambiguous). Choice D is wrong.

Q17: D Non-Oxide minerals: Silicates (14 + 9 + 6 = 29) + Carbonates (7 + 18 + 3 = 28) = 57. Among these, Monoclinic or Orthorhombic: Silicates (14 + 9 = 23) + Carbonates (7 + 18 = 25) = 48. Probability = 48/57 = 16/19.

Q18: 143 Square both sides of sqrt(k - x) = 36 - x: k - x = (36 - x)^2 = 1296 - 72x + x^2. So k = x^2 - 71x + 1296. For exactly one real solution, the minimum value of k occurs at the vertex: x = 71/2 = 35.5. Minimum k = (35.5)^2 - 71(35.5) + 1296 = 1260.25 - 2520.5 + 1296 = 35.75 = 143/4. Therefore 4k = 143. (Verify: at x = 35.5, sqrt(143/4 - 35.5) = sqrt(0.25) = 0.5, and 36 - 35.5 = 0.5.)

Q19: B If (x + 5b) is a factor, substituting x = -5b must yield zero. Choice B: 4(-5b)^2 + 24(-5b) + 20b = 100b^2 - 120b + 20b = 100b^2 - 100b = 100b(b - 1) = 0 when b = 1. Verify: 4x^2 + 24x + 20 = 4(x^2 + 6x + 5) = 4(x + 1)(x + 5), and x + 5 = x + 5(1) = x + 5b with b = 1. Choices A, C, and D yield no positive integer b.

Q20: 40 Rewrite: kx^2 - 54x + 18 = 0. For two distinct real solutions, the discriminant must be positive: (-54)^2 - 4(k)(18) > 0, so 2916 - 72k > 0, giving k < 40.5. The greatest integer value is k = 40.

Q21: 549 Using elimination on 17x - 4y = 31 and 3x + 11y = 52: Multiply the first by 11: 187x - 44y = 341. Multiply the second by 4: 12x + 44y = 208. Add: 199x = 549, so x = 549/199. Therefore p = 549.

Q22: 7.5 Mass of P = 450% of Q = 4.5Q. Mass of P = 0.060% of R = 0.0006R. Setting equal: 4.5Q = 0.0006R, so R = 4.5Q / 0.0006 = 7,500Q. Therefore R is 7,500% of Q, meaning s = 7,500. And s/1,000 = 7.5.

Total Questions: 147 (27 + 27 + 27 + 22 + 22 + 22)